MLX91804 Application note

Library functions' execution information

Contents

1. Scope

The Application Note gives an estimation of the execution time and the stack usage of the main functions included into the prebuilt MLX91804 SW library Rev 1.10.11.

2. Conventions and remarks

All the functions specified in this document correspond to the prebuilt MLX91804 SW library Rev 1.10.11 only. Other library versions can include the functions with different performances.

The specified functions are approximately prioritized for a typical application by 3 levels: High, Middle and Low.

The deprecated functions as well as some special library functions are not included in this document. The special functions supporting SW I2C serial communication, LF transmitting, hardware diagnostic interface and the acceleration measurements with the hardware filtering are not specified in the application note.

2.1. Execution time

The functions' execution time is specified either in periods of the High Frequency Oscillator clock (HFO clk) or directly in time expressed in microseconds (µs) or milliseconds (ms).

In order to convert the "HFO clk" units to the real time, one can just multiply it by the HFO period (e.g. 0.125 µs for the typical 8 MHz clock frequency).

Important!
The specified time always corresponds to the functions' body included into the library. The time needed to execute a call, to pass possible input parameters and to copy the output parameter to the right place depends on the application code and the compilation options and hence it is not included to the specified time. It is a customer's responsibility to verify own binary code and to take into account the additional time above.

For example, depending on the map of the linked modules the function call can be realized in the application either by the short or far MLX16 MCU call instruction that will create a difference in 2 HFO clk of the execution time. Two more HFO clk can be added if the call is not in queue of the MCU. As well, there are many options how the compiler can generate a code passing the parameters to and from the called function.

The inline and macro functions insert own body directly to the application code and hence they don't require any call instruction there. All such cases are marked in the column "Comments" in the tables below.

The execution time can be specified within some limits because it can deviate due to:

- the input parameters directly pointing to the duration (e.g. in the function delay_nops, DEBUG_PIN_PULSE, etc. where the time is specified via the formulas based on the input value);
- the different execution branches;
- the internal function's calls and jumps instructions that can be implemented by the linker in their short or far forms;
- the hardware related events (in this case the execution time is always specified in time units);
- HFO frequency deviation over the device operating range (if the execution time is specified in time units).

For the specification done in time units, the typical, minimal and maximal values correspond to HFO frequency 8, 10.4 and 5.2 MHz respectively. All such values are based on the measurements done at VDD = 3.0 Volt and temperature 25°C and hence cannot be considered as absolute limits over the full operating range.

2.2. Stack usage

The stack usage is specified by taking into account how the function is called. If the function needs MLX16 call instruction in the application code, two bytes are added to the total stack usage. Otherwise, for the inline and macro functions this component results zero. If the application code needs to use the stack to pass the input parameters to the function, the corresponding value is also added to the total stack usage specified for the function.

Please note, the compiler can optimize the application code by different ways, and it can make a difference in the stack usage. For example, if 2 bytes of the stack are taken to pass an input parameter to some function, those 2 bytes can be released later in the application code (not always just after the return from the function). The compiler can release them even after some calls of other functions that will increase their effective stack usage.

3. Library functions

The library functions are specified in groups separated according to the functional meaning.

3.1. Data acquisition

The functions below support AD conversions.

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Adc_ClearAllErrors	Middle	2		21		HFO clk	
Adc_IsError	Middle	2	43	-	46	HFO clk	
Adc_ReadGpio	Low	8	-	145	240	µs	
Adc_ReadPressure	Middle	8	-	165	270	µs	
Adc_ReadPVT	High	10	-	320	530	µs	
Adc_ReadTemperature	Middle	10	-	210	345	µs	
Adc_ReadVoltage	High	10	-	215	350	µs	
low_power_adc_read_pressure	Low	6	-	80	135	µs	
Phys_AccelerationX	High	16	-	0.48	0.8	ms	Specified for the CA versions of device
Phys_AccelerationZ	High	16	-	0.48	0.8	ms	
Accel_GToRawX/Z	Middle	6	92	-	102	HFO clk	
Accel_RawToGX/Z	Middle	6	67	-	78	HFO clk	

Note: The function name Accel_GToRawX/Z has a deprecated alias Accel_ToAduX/Z.

3.2. Calculation SW module

The calculation SW module includes the functions implementing the measurements' compensating and a computation of the 24-bit signature based on the ROM BIST hardware.

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Calc_RomBistFor(a0, a1) [a0…a1] belongs to Flash	Low	8	0.77•T	T= 21.5 + N	1.54•T	µs	1) N is a number of the 16-bit words covered by the ROM BIST. N = (a1-a0)>>1, where a0 and a1 are the input parameters defining the memory address range covered by the signature calculation 2) The execution time for a failure case is not specified 3) Stack usage includes 2 bytes used to pass the input parameter
Calc_RomBistFor(a0, a1) [a0…a1] is out of Flash	Low	8	0.77•T	T= 21 + 0.25•N	1.54•T	µs	
Calc_Pressure	High	34	-	100	165	µs	
Calc_Temperature	High	12	174	-	193	HFO clk	
Calc_Voltage	High	2	36	-	37	HFO clk	

3.3. Temperature sensor

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
TempSensor_GetMode	Low	2		21		HFO clk	
TempSensor_GetStatus	Low	2	38	-	39	HFO clk	
TempSensor_IsInRange	Low	2	45	-	49	HFO clk	
TempSensor_SetMode	Low	2		16		HFO clk	

3.4. LF transceiver

The functions below are used to support the LF receiver of the LF transceiver.

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Lf_Disable	High	2		21		HFO clk	
Lf_GetState	High	2	33	-	34	HFO clk	
Lf_Init	High	10	444	-	446	HFO clk	
Lf_Start	High	2		26		HFO clk	
LfRx_FifoCount	High	2		21		HFO clk	
LfRx_FifoFlush	High	2		21		HFO clk	
LfRx_FifoPeekByte	High	2		21		HFO clk	
LfRx_FifoReadByte	High	2		30		HFO clk	
LfRx_IsFifoFull	High	2	28	-	31	HFO clk	
LfRx_IsFifoOverrun	High	2	47	-	50	HFO clk	
LfRx_IsFifoUnderrun	High	2	47	-	50	HFO clk	
LfRx_ReceivedHeader	High	2		25		HFO clk	

3.5. RF transmitter

Function name	Importance in TPMS application	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
RfTx_Calibrate	High	38	-	1.7	4	ms	The typical value corresponds to about 6 measurement cycles of the search algorithm and is specified for the CA versions of the device; the number of search cycles strongly depends on the applied temperature
Radio_Disable	High	2		21		HFO clk	
RfTx_FifoCount	High	2	21	-	22	HFO clk	
RfTx_FifoFlush	High	2		21		HFO clk	
RfTx_FifoWriteByte	High	2		16		HFO clk	
RfTx_GetStatus	High	2		16		HFO clk	
RfTx_Init	High	16	650	-	660	HFO clk	For the configuration without .extra section
			840	-	850	HFO clk	For the configuration with .extra section
RfTx_IsFifoFull	High	2	28	-	31	HFO clk	
RfTx_IsFifoOverrun	High	2	47	-	50	HFO clk	
RfTx_IsFifoUnderrun	High	2	47	-	50	HFO clk	
RfTx_Run	High	16	-	1.3	2.4	ms	1) The RF transmission time is not included 2) RFTX_DONE and RFTX_HALF interrupts don't contribute to the stack usage 3) Stack usage includes 2 bytes used to pass the input parameter 3) The maximal execution time includes the maximal start-up time of the crystal oscillator 4) The execution time for the failure cases is not specified 5) The typical values are specified for the CA versions of the device
RfTx_RunFromBuf, payload < 33 bytes	Low	24	-	1.3	2.4	ms	
RfTx_RunFromBuf, payload > 32 bytes			-	1.6	2.9	ms	

3.6. HFO and LFO calibrations

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Sys_CalibrateHfo	High	32	2	-	3.3	ms	The maximal execution time includes the maximal start-up time of the crystal oscillator
Sys_CalibrateLfo	High	32	12	-	26	ms	
Sys_CalibrateUlpo	High	32	36	-	47	ms	
Sys_CalibrateHfoByExtClk	Low	32	1	-	1.5	ms	
Sys_CalibrateLfoByExtClk	Low	32	11	-	24	ms	
Sys_CalibrateUlpoByExtClk	Low	32	35	-	46	ms	

Note: The execution time for the failure cases is not specified.

3.7. Timers and delays

Function name	Importance in TPMS application	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
AbsWatchdog_IsWindowOpen	Middle	2	34	-	37	HFO clk	
AbsWatchdog_Restart	Middle	2	12			HFO clk	
SimpleTimer1_GetCounter	Middle	2	12			HFO clk	
SimpleTimer1_Start	Middle	4	56	-	58	HFO clk	Stack usage includes 2 bytes used to pass the input parameter
SimpleTimer1_Stop	Middle	2	24			HFO clk	
SimpleTimer2_GetCounter	Middle	2	12			HFO clk	
SimpleTimer2_Start	Middle	4	56	-	58	HFO clk	Stack usage includes 2 bytes used to pass the input parameter
SimpleTimer2_Stop	Middle	2	24			HFO clk	
WakeupTimer_Init	High	4	74	-	75	HFO clk	Stack usage includes 2 bytes used to pass the input parameter
WakeupTimer_IntClear	High	2	49			HFO clk	
WakeupTimer_Restart	High	2	47			HFO clk	
delay_nops(n), n<256	Middle	0	T = 4 • n + 2			HFO clk	Inline function that doesn't require using a call
delay_nops(n), n>255			T = 4 • n + 4				

3.8. System control

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Sys_GetChipId	High	2	19			HFO clk	
Sys_GetChipIdHi	Low	2	14			HFO clk	
Sys_GetResetReason	Middle	4	71	-	81	HFO clk	
Sys_IsColdBoot	High	2	38	-	41	HFO clk	
Sys_SetNormalLfo	Middle	2	30			HFO clk	
Sys_SetUlpLfo	Middle	2	30			HFO clk	
Sys_ResetCpu	High	2	17	-	19	HFO clk	Reset time is about 4 us
Sys_DeepSleepWith	High	6	75			HFO clk	
Sys_SleepWith	High	6	75			HFO clk	
Sys_StopWith	High	2	36			HFO clk	
Sys_HaltWith	Middle	2	31			HFO clk	

3.9. Atomic sections support

This SW module combines the macros supporting atomic sections with different configurations.

Function name	Importance in TPMS application	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
ENTER_SECTION (ATOMIC_KEEP_MODE)	High	0	25	-	27	HFO clk	These are Macro expressions inserting a code into the application without using call
EXIT_SECTION() after ATOMIC_KEEP_MODE	High	0	10	-	12	HFO clk	
ENTER_SECTION (ATOMIC_SYSTEM_MODE)	High	2	24	-	28	HFO clk	
EXIT_SECTION() after ATOMIC_SYSTEM_MODE	High	0	5	-	7	HFO clk	
ENTER_SECTION (SYSTEM_MODE)	High	2	24	-	28	HFO clk	
EXIT_SECTION() after SYSTEM_MODE	High	0	5	-	7	HFO clk	

Note: The macro EXIT_SECTION generates different code to exit from different atomic sections.

3.10. Wake-Ups and Interrupts

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Itc_ClearPending	High	2		21		HFO clk	
Itc_Disable	High	2	26	-	30	HFO clk	
Itc_Enable	High	2	26	-	30	HFO clk	
Itc_IsPending	High	2	26	-	39	HFO clk	Execution time varies depending on the requested bit's position in the addressing word
Itc_SetPrio	Middle	4	68	-	71	HFO clk	
Wakeup_Disable	High	2	21	-	35	HFO clk	
Wakeup_Enable	High	2	21	-	35	HFO clk	

3.11. General-Purpose IOs (GPIO)

Function name	Importance in TPMS application	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Gpio_Init	Low	2		63		HFO clk	
Gpio_ReadPin(GPIO0)	Low	2	57	-	60	HFO clk	
Gpio_ReadPin(GPIO1)			57	-	60	HFO clk	
Gpio_ReadPin(GPIO2)			72	-	75	HFO clk	
Gpio_ReadPin(GPIO3)			87	-	90	HFO clk	GPIO3 is not accessible externally
Gpio_ReadPin(>3)				66		HFO clk	Corresponds to the wrong GPIO ID
Gpio_ReadPort	Low	2		21		HFO clk	
Gpio_SetPinMode	Low	6	34	-	54	HFO clk	1) Stack usage includes 2 bytes used to pass the input parameter 2) The minimal execution time corresponds to the case with a wrong GPIO ID"
Gpio_TogglePin Toggling from high to low	Low	2	36	-	70	HFO clk	The minimal execution time corresponds to the case with a wrong GPIO ID
Gpio_TogglePin Toggling from low to high					73	HFO clk	
Gpio_WritePin	Low	8	97	-	124	HFO clk	Stack usage includes 2 bytes used to pass the input parameter

3.12. nvRAM and Flash memory

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Flash_ClearSingleErrorCounter	Middle	2	21			HFO clk	
Flash_GetSingleErrorCounter	Middle	2	16			HFO clk	
Flash_PageWrite	Middle	26	-	5.3	9	ms	1) The execution time for a failure case is not specified. 2) Stack usage includes 6 bytes used to pass the input parameter
Flash_SectorErase	Middle	54	-	180	300	ms	1) The execution time for a failure case is not specified. 2) Stack usage includes 4 bytes used to pass the input parameter
NvRam_IsDoubleErrorDetected	Low	2	24	-	27	HFO clk	
NvRam_ClearDoubleError	Low	2	17	-	18	HFO clk	
NvRam_IsSingleErrorCorrected	Middle	2	24	-	27	HFO clk	
NvRam_ClearSingleError	Middle	2	17	-	18	HFO clk	
NvRam_RestoreHfoTrimming	Middle	2	23			HFO clk	Since the function changes the HFO configuration, clock periods can vary during the execution time
NvRam_RestoreLfoTrimming	Middle	2	23			HFO clk	
NvRam_RestoreOscTrimming	Low	2	35			HFO clk	Since the function changes the HFO configuration, clock periods can vary during the execution time
NvRam_RestoreUlpoTrimming	Middle	2	23			HFO clk	
NvRam_Recall	High	4 40	50 130	70 185	105 285	µs µs	VDD > 2.4V VDD < 2.4V
NvRam_Store	High	2	-	11	13	ms	

3.13. Debug SW module

The Debug SW module can be used to monitor the firmware execution on its design stage.

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
DEBUG_ACK_AWD	Middle	0	10	-	12	HFO clk	Inline function that doesn't require a call
DEBUG_PIN_INIT(LOGIC_HIGH)	Middle	2	38			HFO clk	
DEBUG_PIN_INIT(LOGIC_LOW)			33			HFO clk	
DEBUG_PIN_PULSE(n)	Middle	2	0.77•T	T = 3.5 + 19.7•n	1.54•T	µs	
DEBUG_PIN_TOGGLE(n)	Middle	2	0.77•T	T = 1.24 + 4.36•n	1.54•T	µs	
DEBUG_PUT_BYTE	Middle	4	-	100	105	µs	Specified for the HFO frequency: 8 MHz ± 1%
DEBUG_PUT_STRING with n symbols	Middle	8	0.77•T	T = 4.4 + 108•n	1.54•T	µs	
DEBUG_PUT_WORD	Middle	8	-	205	210	µs	
DEBUG_PUT_WORD_HEX	Middle	16	-	0.47	0.7	ms	

Note:
1) Because the functions DEBUG_PUT_BYTE, DEBUG_PUT_STRING, DEBUG_PUT_WORD and DEBUG_PUT_WORD_HEX use the SW UART to output the debug data, it requires an accurate HFO frequency 8 MHz to keep the duration of the output bits within the acceptable limits. That is why, the functions execution time is specified for the HFO frequency 8 MHz ± 1% that is achievable after the HFO calibration by the function Sys_CalibrateHfo.
2) The output format and the corresponding execution time of the function DEBUG_PUT_WORD_HEX can vary depending on the output data value.

3.14. System validation

Function name	Importance in TPMS app	Stack usage	Execution time				Comments
			Min	Typ	Max	Unit	
Check_IsBadMlxData	L	14	80	120	200	us	
Check_IsBadAccelX	H	18	-	0.85	1.4	ms	Stack usage includes 2 bytes used to pass the input parameter
Check_IsBadAccelZ	H	18	-	0.85	1.4	ms	Stack usage includes 2 bytes used to pass the input parameter
Check_IsBadPressure	H	4	-	156	250	us	

Note: The execution time for the failure cases is not specified.

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